# Garrett Smith

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Mechanical Engineering junior with experience in CAD/DFM, design reviews, and test-rig integration, focused on mechanical design, rapid prototyping, and system integration. Interested in robotics automation and mechatronic systems.

## EDUCATION

**Cornell University,** College of Engineering, Ithaca, NY **Expected May 2027**
Bachelor of Science, **Mechanical Engineering**, GPA: 3.85 Minor: Environment and Sustainability

***Relevant Coursework:*** Thermodynamics • Dynamics • Mechanical Synthesis • Fluid Mechanics • System Dynamics • Mechanics of Engineering Materials • Heat Transfer • Mechatronics • Machine Learning

## SPECIALIZED SKILLS

**Technical:** Autodesk Fusion 360, SolidWorks, DraftSight, MATLAB + Simulink, Ansys, NetSuite, Python/Machine Learning, Decision Modeling

## ENGINEERING EXPERIENCE

**Solar Panel Reboot Team,** *Cornell University Sustainable Design,* Ithaca, NY **Jan 2024 – Present**

**Co-Team Lead,** Jan 2026 – Present

- 15+ hours per week directing system-level project architecture, translating high-level goals into actionable technical requirements and working closely with all 17 members to ensure 100% on-time delivery of project milestones
- Orchestrated rebuild of our indoor solar-simulator testing rig, streamlining the integration of mechanical and electrical subsystems and fostering a culture of rapid iteration, quantitative analysis, and safety checks
- Managed external collaboration with Corning Inc., pushing forward our research project analyzing the effectiveness of hydrophobic and hydrophilic coatings in reducing soiling effects for use in urban environments

**Subteam Member,** Jan 2024 – Jan 2026

- Ran initial design and assembly of solar simulator, allowing data-driven comparisons of refurbished PV performance
- Contributed to anti-soiling coating evaluation through literature reviews and experimental planning support

**Hardware Engineering Intern,** *Advanced Environmental Monitoring*, Hayward, CA (Remote) **May – Aug 2025**

- Built parametric CAD models and led design reviews to validate fit across multiple solar panel sizes and poles
- Led solar add-on design review, including ASCE 7 wind calcs, fastener sizing, and coastal corrosion plan
- Performed item-master analysis across 117 solar panel SKUs, recommending consolidation to 9 "A-Tier" SKUs through a weighted-factor decision matrix for uniformity and consistency across 4 different business units
- Standardized DraftSight templates, saving design team time and improving drawing consistency

## PROJECTS

**OpenUP,** Split Water Bottle, *Cornell University* **Feb 2025 – May 2025**

- Utilized a user-centric design strategy, focused on mitigating bacteria growth in traditional bottles, integrated DfX and Pugh matrices to scope product around easy cleanability and leakproof sealing and balance constraints
- Accelerated product development cycle -› from 30+ initial concepts to 6 fully defined CAD models and using two rapid prototyping iterations to validate fit, finish, and mechanical alignment for a one-of-a-kind split water bottle
- Engineered a 100% leakproof sealing system by switching from epoxy to heat-set inserts for clamp attachment and optimizing the tradeoff between seal efficiency and ease of use through O-ring compression calculations
- Developed a comprehensive industrialization roadmap to transition 3D printed prototypes to a commercial grade product, using stainless steel deep-drawing, thread rolling, and spot-welding for durability and DFQ standards

## ADDITIONAL EXPERIENCE

**Construction and Maintenance Employee,** *Mike Smith Contracting*, McVeytown, PA **May – Aug 2024**

- Assisted with house flipping projects, gaining hands-on experience in construction and project management
- Adapted to a dynamic work environment and demonstrated my ability to effectively learn a new set of skills while exemplifying reliability and technical competence as I contributed to achieving desired project goals and successes

**Independent Tutoring,** *Cornell University* **Sept 2023 – Present**

- Provided free student tutoring services to fellow students on the mechanical engineering track, providing 10+ hours per week helping classmates and have received consistent feedback regarding outcome improvements